EXP-1

VERIFICATION OF KIRCHHOFF'S LAWS

AIM

a. To verify Kirchhoff's Voltage Law (KVL) for the given circuit.

b. To verify Kirchhoff's Current Law (KCL) for the given circuits.

APPARATUS REQUIRED:

The apparatus required for the experiment are three 1 kΩ resistors, three DC voltmeters with a range of 0–30 V, three DC ammeters with a range of 0–200 mA, one breadboard, one regulated power supply with a range of 0–30 V, and connecting wires as required.

THEORY:

KVL: Kirchhoff's voltage law states that the sum of the voltage differences around any closed loop in a circuit must be zero. A loop in a circuit is any path that ends at the same point at which it starts.

KCL: Kirchhoff's Current Law (KCL) Kirchhoff's Current Law states that the algebraic sum of the currents entering and leaving a node is equal to zero. By convention, currents entering the node are positive, and those leaving a node are negative

PROCEDURE:

a. **KVL**:

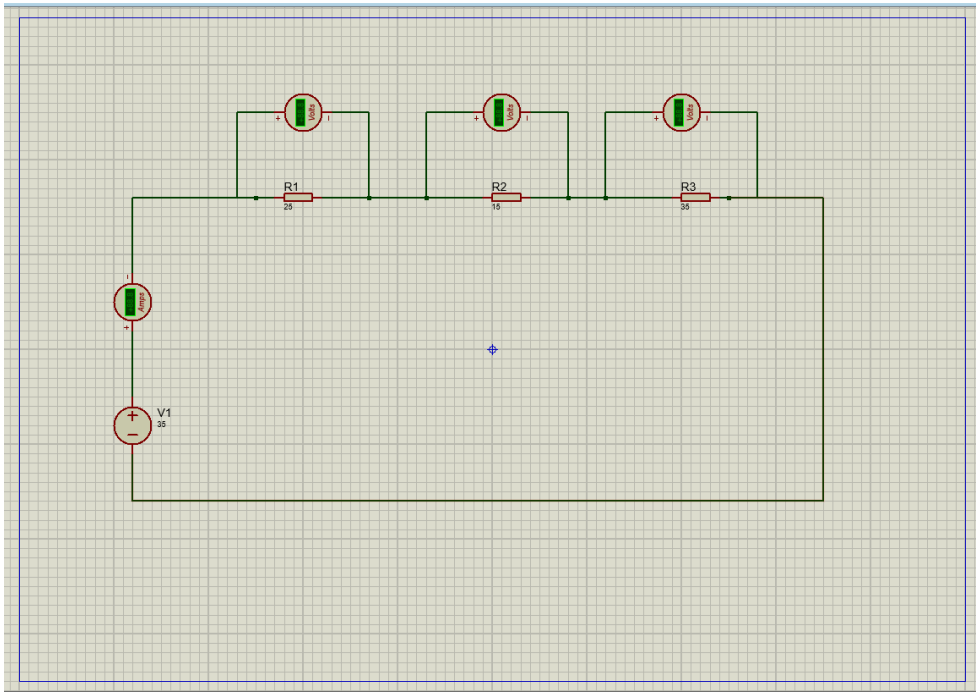
1. Connect as per the circuit diagram.
2. Check if the RPS voltage is set to zero voltage.
3. Check all the meters for null position.
4. Switch on the RPS.
5. Set the input voltage to a value between 0V to 30V.
6. Record the voltage values shown in the voltmeter connected across each resistor.
7. Take readings for different values of input voltage and tabulate them.

b. **KCL**:

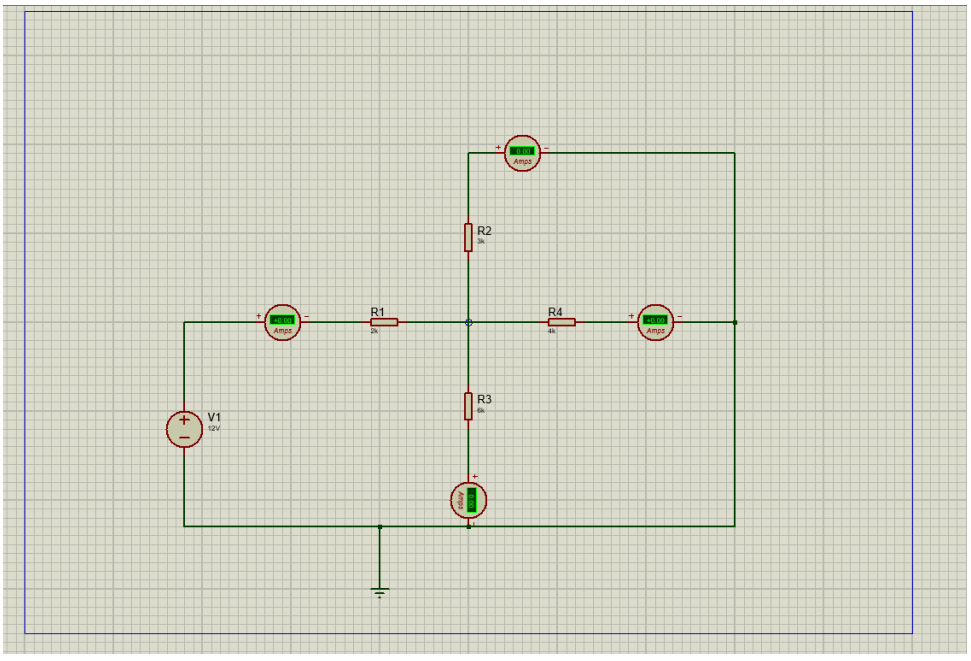
1. Connect as per the circuit diagram.
2. Check if the RPS voltage is set to zero voltage.
3. Check all the meters for null position.
4. Switch on the RPS.
5. Set the input voltage to a value between 0V to 30V.
6. Record the voltage values shown in the ammeter connected to each resistor.
7. Take readings for different values of input voltage and tabulate them.

CIRCUIT DIAGRAM:

a. KVL:



b. KCL:



Calculation:

a. KVL:

A. KVL CALCULATION:

$V = 35V$
$R_1 = 25\Omega$
$R_2 = 15\Omega$
$R_3 = 35\Omega$

All three resistors are connected in series

$R_T = R_1 + R_2 + R_3$
$R_T = 25 + 15 + 35$
$\boxed{R_T = 75\Omega}$

$$I = \frac{V}{R_T}$$
$$I = \frac{35}{75}, \quad I = 0.4667A$$
$$\boxed{I \approx 0.47A}$$

$V_1 = IR_1$
$V_1 = 0.4667 \times 25$
$V_1 = 11.667V \Rightarrow \boxed{V_1 = 11.67V}$

$V_2 = IR_2$
$V_2 = 0.4667 \times 15$
$V_2 = 7.0005V$
$\boxed{V_2 = 7.00V}$

$V_3 = IR_3$
$V_3 = 0.4667 \times 35$
$\boxed{V_3 = 16.33V}$

Verification of KVL:

$V = V_1 + V_2 + V_3$
$35 = 11.67 + 7.00 + 16.33$
$35 = 35.00V$
$\boxed{V = V_1 + V_2 + V_3}$ $\boxed{35V = 35V}$

∴ KVL is verified //.

b. KCL:

KCL CALCULATION:

Given:

$V = 12V$

$R_1 = 2K\Omega, R_2 = 3K\Omega, R_3 = 6K\Omega, R_4 = 4K\Omega$

Since R_2, R_3, R_4 are in parallel,

$$\frac{1}{R_p} = \frac{1}{3} + \frac{1}{6} + \frac{1}{4}$$

$$R_p = 1.333K\Omega$$

Total resistance:

$$R_T = R_1 + R_p$$

$$R_T = 2 + 1.333 = 3.333K\Omega$$

$$I = \frac{12}{3.333K} = 3.6mA$$

Voltage across parallel branches:

$$V_p = 12 - (3.6mA \times 2K\Omega)$$

$$V_p = 12 - 7.2 = 4.8V$$

$$I_2 = \frac{4.8}{3K} = 1.6mA$$

$$I_3 = \frac{4.8}{6K} = 0.8mA, \quad I_4 = \frac{4.8}{4K} = 1.2mA$$

According to KCL,

$$I = I_2 + I_3 + I_4$$

$$36 = 1.6 + 0.8 + 1.2$$

$$3.6mA = 3.6mA$$

Hence KCL verified.

Tabulation:

a. KVL:

Quantity	Calculated value
Supply voltage V	35V
Circuit current I	0.47 A
VR1	11.67 V
VR2	7.00 V
VR3	16.33 V
Sum of voltage drops	35.00 V

b. KCL:

Quantity	Calculated value
Supply voltage	1 V
Current entering junction	75 μA
Current through	25 μA
Current through	25 μA
Current through	25 μA
Total current leaving	75 μA

RESULT:

Thus, for the given circuit, Kirchhoff’s Laws, (a) KVL and (b) KCL are proved.